

Traversable wormholes in $f(T, \tau)$ gravity: a complete classification of the non-exotic sector

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We study static and spherically symmetric traversable wormholes in $f(T, \tau)$ gravity, where the torsion scalar T is coupled to the trace τ of the matter energy-momentum tensor. We consider the linear model $f(T, \tau) = T + \beta\tau$ with an anisotropic fluid and the mean-pressure matter Lagrangian $\mathcal{L}_m = \mathcal{P} = (p_r + 2p_t)/3$. The field equations are obtained for the Morris–Thorne geometry without fixing the redshift or shape function at the outset. For a constant redshift function, the energy-condition problem takes a simple form. On the branch $\beta > 8\pi$ and for $b(r) > 0$, the energy density together with the null, weak, and strong energy conditions is satisfied throughout the spacetime if and only if $rb(r)$ is non-increasing. The same condition also implies asymptotic flatness, $b(r) < r$ outside the throat, and $b'(r_0) \leq -1$. The allowed geometries can therefore be written as $b(r) = r_0^2 h(r)/r$, where $h(r_0) = 1$ and $h(r)$ is positive and non-increasing. For the representative family $b(r) = r_0(r_0/r)^n$, the null, weak, and strong energy conditions hold for $n \geq 1$, while the dominant energy condition requires $n \geq 3(\beta - 2\pi)/(\beta - 6\pi)$. We also separate the physical matter from the effective source and show how the trace coupling allows the physical matter to remain non-exotic although the effective source violates the null energy condition. Finally, we examine the marginal case $b(r) = r_0^2/r$ with a non-constant redshift function. A decreasing redshift function can improve the tangential null energy condition at the throat, but this improvement cannot be maintained throughout an asymptotically flat exterior. These results show that the matter–torsion coupling can support a broad class of traversable wormholes without requiring exotic physical matter.

I. INTRODUCTION

Wormholes are non-trivial spacetime geometries that connect two distant regions through a throat. Their early history goes back to the Einstein–Rosen construction [1], while the modern study of traversable wormholes was established by Morris and Thorne [2, 3]. Several simple and thin-shell constructions were subsequently developed [4–6], and standard reviews can be found in Refs. [7, 8].

The main difficulty in general relativity is that the flare-out condition at the throat requires a violation of the null energy condition (NEC). The supporting matter is therefore exotic in the usual general-relativistic interpretation. Considerable effort has been devoted to reducing the amount of exotic matter or confining it to a limited region [9, 10]. Wormholes supported by phantom or Chaplygin-type matter have been studied in Refs. [11–16], while the Casimir effect provides another interesting source of negative energy [17]. Stability and junction constructions have also been widely investigated [12, 18, 19].

Modified gravity offers a different way to address the same problem. The additional geometrical terms can contribute to the effective source that supports the throat, allowing the physical matter sector to satisfy the standard energy conditions in suitable parameter ranges. This possibility has been explored in viable $f(R)$ gravity [20], Einstein–Gauss–Bonnet gravity [21], extra-dimensional scenarios [22], $f(Q)$ gravity [23], Rastall gravity [24–27], and $f(R, \mathcal{L}_m)$ gravity [28, 29].

Matter–geometry coupling has attracted particular attention in $f(R, T)$ gravity [30]. A wide variety of wormhole models have been constructed in this framework, including static and Morris–Thorne solutions, charged configurations, conformally symmetric geometries, and models supported without exotic matter [31–42]. More recently, variable equations of state have also been considered [43]. These studies show that a direct coupling between matter and geometry can substantially change the material requirements for a traversable throat.

Teleparallel gravity provides an alternative description of gravitation in which torsion, rather than curvature, carries the gravitational interaction. Its extensions based on a general function $f(T)$ have been studied in a wide range of cosmological and astrophysical settings [44–48]. Wormhole geometries in modified teleparallel gravity were analysed in Ref. [49], where the role of the energy conditions was examined explicitly.

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A further extension is obtained by coupling the torsion scalar to the trace of the matter energy–momentum tensor. The resulting $f(T, \mathcal{T})$ theory was introduced by Harko *et al.* [50]. In the present work we denote the trace by τ to avoid confusing it with the torsion scalar T . The theory has been applied to cosmology and dynamical systems [51–57], brane configurations [58], gravastars [59], and compact stars [60–65]. Black-hole thermodynamics in related matter-coupled teleparallel models has also been discussed in Ref. [66].

Wormholes in $f(T, \tau)$ gravity have received increasing attention. The influence of GUP-corrected Casimir energy on zero-tidal-force wormholes was studied in Ref. [67]. Embedding class-I solutions based on the Karmarkar condition were considered in Ref. [68], and a systematic analysis of wormholes in the linear theory was recently presented in Ref. [69]. In that study, particular matter relations, including linear and variable equations of state, were imposed to obtain explicit wormhole geometries. Our approach reverses this logic: no equation of state is assumed, and the energy conditions themselves are used to classify the admissible geometry. This leads to a necessary-and-sufficient condition for the constant-redshift sector, the general class $b(r) = r_0^2 h(r)/r$, and a global obstruction for the marginal member when the redshift function is allowed to vary. These results make $f(T, \tau)$ gravity a natural setting in which to ask whether the matter threading a wormhole can satisfy the standard energy conditions.

The purpose of the present work is to approach this question without first choosing an equation of state or imposing a particular shape function. We consider the linear model

$$f(T, \tau) = T + \beta\tau, \quad (1)$$

with an anisotropic matter source and the mean-pressure matter Lagrangian $\mathcal{L}_m = (p_r + 2p_t)/3$. We first derive the field equations for a general Morris–Thorne geometry. In the constant-redshift sector, the energy conditions themselves select the allowed shape functions. We show that for $\beta > 8\pi$ the simple condition $(rb)' \leq 0$ determines a complete class of asymptotically flat wormholes supported by non-exotic physical matter. No equation of state or traceless-matter condition is required. We then examine the marginal member $b(r) = r_0^2/r$ and ask whether a non-constant redshift function can improve its tangential null-energy behaviour away from the throat.

The paper is organised as follows. In Sec. II we introduce the $f(T, \tau)$ model and derive the wormhole field equations. The constant-redshift solutions are studied in Sec. III. In Sec. IV we examine the marginal shape function with a non-constant redshift. The main results are summarised in Sec. V. Throughout the paper we use

geometrised units $G = c = 1$ and the metric signature $(-, +, +, +)$.

II. $f(T, \tau)$ GRAVITY AND WORMHOLE FIELD EQUATIONS

A. Basic formulation

Teleparallel gravity is formulated in terms of a tetrad $e^A{}_\mu$, from which the metric is constructed as

$$g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu, \quad e \equiv \det(e^A{}_\mu) = \sqrt{-g}, \quad (2)$$

with $\eta_{AB} = \text{diag}(-1, 1, 1, 1)$. In the pure-tetrad representation, $\omega^A{}_{B\mu} = 0$, and the torsion tensor is

$$T^\lambda{}_{\mu\nu} = e_A{}^\lambda (\partial_\mu e^A{}_\nu - \partial_\nu e^A{}_\mu). \quad (3)$$

The contorsion tensor and superpotential are

$$K^{\mu\nu}{}_\rho = -\frac{1}{2} (T^{\mu\nu}{}_\rho - T^{\nu\mu}{}_\rho - T_\rho{}^{\mu\nu}), \quad (4)$$

$$S_\rho{}^{\mu\nu} = \frac{1}{2} (K^{\mu\nu}{}_\rho + \delta_\rho{}^\mu T^{\alpha\nu}{}_\alpha - \delta_\rho{}^\nu T^{\alpha\mu}{}_\alpha), \quad (5)$$

so that the torsion scalar is

$$T = S_\rho{}^{\mu\nu} T^\rho{}_{\mu\nu}. \quad (6)$$

The action of $f(T, \tau)$ gravity is given by [50]

$$S = \frac{1}{16\pi} \int d^4x e f(T, \tau) + \int d^4x e \mathcal{L}_m. \quad (7)$$

For the anisotropic source considered here,

$$\Theta^\mu{}_\nu = \text{diag}(-\rho, p_r, p_t, p_t), \quad (8)$$

with trace

$$\tau = -\rho + p_r + 2p_t. \quad (9)$$

We take the matter Lagrangian to be the mean pressure,

$$\mathcal{L}_m = \mathcal{P} \equiv \frac{p_r + 2p_t}{3}, \quad (10)$$

which reduces to $\mathcal{L}_m = p$ in the isotropic limit. As in the standard perfect-fluid treatment of trace-coupled models [50], we adopt the closure prescription $\partial^2 \mathcal{P} / \partial g^{\mu\nu} \partial g^{\alpha\beta} = 0$. The corresponding matter variation is then

$$\Theta_{\mu\nu}^{(\text{var})} = -2\Theta_{\mu\nu} + \mathcal{P} g_{\mu\nu}. \quad (11)$$

This is a prescription for the anisotropic matter Lagrangian rather than a microscopic statement about every anisotropic fluid.

Variation of Eq. (7) with respect to the tetrad gives

$$e_A{}^\rho S_\rho{}^{\mu\nu}(\partial_\mu T)f_{TT} + e_A{}^\rho S_\rho{}^{\mu\nu}(\partial_\mu \tau)f_{T\tau} + e^{-1}\partial_\mu(ee_A{}^\rho S_\rho{}^{\mu\nu})f_T + e_A{}^\mu T^\lambda{}_{\mu\kappa}S_\lambda{}^{\nu\kappa}f_T - \frac{1}{4}e_A{}^\nu f - \frac{f_\tau}{2}(e_A{}^\lambda \Theta_\lambda{}^\nu + \mathcal{P}e_A{}^\nu) = -4\pi e_A{}^\lambda \Theta_\lambda{}^\nu. \quad (12)$$

For the linear model in Eq. (1), $f_T = 1$, $f_\tau = \beta$, and $f_{TT} = f_{T\tau} = 0$. The field equations reduce to

$$G^\mu{}_\nu = (8\pi - \beta)\Theta^\mu{}_\nu - \beta\left(\mathcal{P} + \frac{\tau}{2}\right)\delta^\mu{}_\nu, \quad (13)$$

where

$$\mathcal{P} + \frac{\tau}{2} = \frac{-3\rho + 5p_r + 10p_t}{6}. \quad (14)$$

The general-relativistic equations are recovered continuously in the limit $\beta \rightarrow 0$. For this linear choice the geometric sector isTEGR, while the modification enters through the trace-coupled source. The structure is therefore analogous to, but not identical with, the linear $f(R, T)$ case [30], because the matter variation leads to the specific coefficients appearing in Eq. (13).

The physical matter tensor is not separately conserved when $\beta \neq 0$. Taking the covariant derivative of Eq. (13) and using the Bianchi identity gives

$$\nabla_\mu \Theta^\mu{}_\nu = \frac{\beta}{8\pi - \beta} \partial_\nu \left(\mathcal{P} + \frac{\tau}{2}\right). \quad (15)$$

Thus the trace coupling describes an exchange of energy-momentum between the physical matter and the gravitational sector, as is typical of matter-geometry coupled theories.

The diagonal tetrad is consistent for this linear model. For a static and spherically symmetric spacetime, the off-diagonal equation reduces to

$$f_{TT}T' + f_{T\tau}\tau' = 0, \quad (16)$$

which is identically satisfied because $f_{TT} = f_{T\tau} = 0$. In the pure-tetrad representation the diagonal spherical

tetrad carries the familiar inertial torsion contribution even in the Minkowski limit, but this produces no additional field-equation constraint for the present linear model precisely because $f_{TT} = f_{T\tau} = 0$.

B. Wormhole geometry and field equations

We consider the Morris–Thorne line element

$$ds^2 = -e^{2\Phi(r)}dt^2 + \frac{dr^2}{1 - b(r)/r} + r^2(d\theta^2 + \sin^2\theta d\varphi^2), \quad (17)$$

where $\Phi(r)$ and $b(r)$ are the redshift and shape functions, respectively. A throat at $r = r_0$ satisfies

$$b(r_0) = r_0, \quad b'(r_0) < 1, \quad b(r) < r \quad (r > r_0), \quad (18)$$

and $\Phi(r)$ must remain finite to avoid a horizon. Asymptotic flatness requires $b(r)/r \rightarrow 0$ and $\Phi(r) \rightarrow \text{const}$ as $r \rightarrow \infty$.

A diagonal tetrad compatible with Eq. (17) is

$$e^A{}_\mu = \text{diag}\left(e^\Phi, \frac{1}{\sqrt{1 - b/r}}, r, r \sin\theta\right), \quad (19)$$

which gives

$$T(r) = -\frac{2}{r}\left(1 - \frac{b}{r}\right)\left(2\Phi' + \frac{1}{r}\right). \quad (20)$$

Substituting the metric into Eq. (13) and solving for the anisotropic source yields

$$\rho = \frac{1}{Dr^2}\left[-10\beta r(r - b)\Phi'^2 + 5\beta\Phi'\{3b + r(b' - 4)\} - 16(\beta - 3\pi)b' - 10\beta r(r - b)\Phi''\right], \quad (21)$$

$$p_r = \frac{1}{Dr^3}\left[10\beta r^3\Phi'^2 - 8\beta r b' + r^2\Phi'(96\pi - 28\beta - 5\beta b') + 10\beta r^3\Phi'' - b\{48\pi - 24\beta + (96\pi - 33\beta)r\Phi' + 10\beta r^2\Phi'^2 + 10\beta r^2\Phi''\}\right], \quad (22)$$

$$p_t = \frac{1}{Dr^3}\left[b\{-12(\beta - 2\pi) - 3(\beta + 8\pi)r\Phi' + 2(7\beta - 24\pi)r^2(\Phi'^2 + \Phi'')\} + r\{-2(7\beta - 24\pi)r^2(\Phi'^2 + \Phi'') + 4(\beta - 6\pi)b' + r\Phi'[48\pi - 4\beta + (7\beta - 24\pi)b']\}\right], \quad (23)$$

where

$$D = 24(\beta - 8\pi)(\beta - 2\pi). \quad (24)$$

These expressions determine the matter source for any

Morris–Thorne geometry in the linear theory, provided $\beta \neq 2\pi, 8\pi$.

The combinations entering the NEC are much simpler:

$$\rho + p_r = \frac{b(1 + 2r\Phi') - r(2r\Phi' + b')}{(\beta - 8\pi)r^3}, \quad (25)$$

$$\rho + p_t = \frac{1}{2(\beta - 8\pi)r^3} \left[b\{-1 + r\Phi' + 2r^2\Phi'^2 + 2r^2\Phi''\} - r\{2r^2\Phi'^2 - r\Phi'(b' - 2) + b' + 2r^2\Phi''\} \right]. \quad (26)$$

The trace-dependent term in Eq. (13) is proportional to δ^μ_ν and therefore cancels when two diagonal components are subtracted. Thus

$$\rho + p_r = \frac{G^t_t - G^r_r}{\beta - 8\pi}, \quad \rho + p_t = \frac{G^t_t - G^\theta_\theta}{\beta - 8\pi}. \quad (27)$$

At the throat, Eq. (25) reduces to

$$(\rho + p_r)|_{r_0} = \frac{1 - b'(r_0)}{(\beta - 8\pi)r_0^2}. \quad (28)$$

The derivative $\Phi'(r_0)$ cancels exactly. Since the flare-out condition gives $1 - b'(r_0) > 0$, the radial NEC can be satisfied at a genuine throat only for

$$\beta > 8\pi. \quad (29)$$

For $\beta < 8\pi$, including the GR limit $\beta \rightarrow 0$, the radial NEC is necessarily violated at the throat. The values $\beta = 2\pi$ and $\beta = 8\pi$ correspond to degeneracies of the algebraic reconstruction and are not included in the following analysis.

III. WORMHOLE SOLUTIONS WITH A CONSTANT REDSHIFT FUNCTION

We now consider the zero-tidal-force case. A finite constant redshift can be absorbed into a redefinition of the time coordinate, so we set $\Phi = 0$. Equations (21)–(23) become

$$\rho = -\frac{2(\beta - 3\pi)b'}{3(\beta - 8\pi)(\beta - 2\pi)r^2}, \quad (30)$$

$$p_r = \frac{3(\beta - 2\pi)b - \beta rb'}{3(\beta - 8\pi)(\beta - 2\pi)r^3}, \quad (31)$$

$$p_t = \frac{-3(\beta - 2\pi)b + (\beta - 6\pi)rb'}{6(\beta - 8\pi)(\beta - 2\pi)r^3}. \quad (32)$$

The trace is

$$\tau = \frac{2(\beta - 6\pi)b'}{3(\beta - 8\pi)(\beta - 2\pi)r^2}. \quad (33)$$

Thus the trace is generally non-zero and the coupling that defines the theory remains active on the solutions considered below.

For an anisotropic fluid, the NEC requires $\rho + p_r \geq 0$ and $\rho + p_t \geq 0$. The weak energy condition (WEC) adds $\rho \geq 0$, the strong energy condition (SEC) adds $\rho + p_r + 2p_t \geq 0$, and the dominant energy condition (DEC) requires $\rho \geq |p_r|$ and $\rho \geq |p_t|$.

A. Energy conditions and admissible shape functions

For constant redshift,

$$\rho + p_r = \frac{b - rb'}{(\beta - 8\pi)r^3}, \quad \rho + p_t = -\frac{b + rb'}{2(\beta - 8\pi)r^3}. \quad (34)$$

We restrict attention to the physically relevant branch $\beta > 8\pi$. Then

$$\rho \geq 0 \iff b' \leq 0, \quad \rho + p_r \geq 0 \iff b' \leq \frac{b}{r}, \quad (35)$$

whereas the tangential NEC gives

$$b + rb' \leq 0 \iff (rb)' \leq 0. \quad (36)$$

For a positive shape function, $b(r) > 0$, this is the strongest condition. It implies $b' \leq -b/r < 0$, so $\rho \geq 0$, while $b - rb' \geq 2b > 0$ guarantees the radial NEC. The SEC is also non-negative under the same restriction.

Hence, for $\beta > 8\pi$ and constant redshift, a positive Morris–Thorne shape function satisfies the energy density, NEC, WEC, and SEC throughout the exterior if and only if $rb(r)$ is non-increasing.

This condition also fixes the asymptotic behaviour. Integrating $(rb)' \leq 0$ from the throat gives

$$rb(r) \leq r_0^2, \quad (37)$$

so that

$$b(r) \leq \frac{r_0^2}{r}, \quad \frac{b(r)}{r} \leq \left(\frac{r_0}{r}\right)^2. \quad (38)$$

Therefore $b(r)/r \rightarrow 0$ as $r \rightarrow \infty$, $b(r) < r$ for every $r > r_0$, and

$$b'(r_0) \leq -1, \quad (39)$$

which is stronger than the usual flare-out requirement. In this sector, asymptotic flatness is not an additional assumption: it follows from the requirement that the physical matter be non-exotic. A further consequence is that $b(r) \rightarrow 0$ at spatial infinity. Hence the ADM mass of every geometry in this admissible constant-redshift class vanishes,

$$M_{\text{ADM}} = \frac{1}{2} \lim_{r \rightarrow \infty} b(r) = 0. \quad (40)$$

The zero asymptotic mass will be consistent with the effective-source picture discussed below, where the positive physical matter contribution is modified by the trace coupling.

The complete class can be written as

$$b(r) = \frac{r_0^2}{r} h(r), \quad (41)$$

with

$$h(r_0) = 1, \quad h(r) > 0, \quad h'(r) \leq 0. \quad (42)$$

Thus the energy conditions determine a family of admissible geometries rather than merely constraining a particular ansatz.

B. Power-law solutions and physical properties

A convenient representative is

$$b(r) = r_0 \left(\frac{r_0}{r} \right)^n, \quad n > 0. \quad (43)$$

For this family,

$$(rb)' = -(n-1) \frac{r_0^{n+1}}{r^n}, \quad (44)$$

so Eq. (36) requires $n \geq 1$. The throat condition is satisfied with $b'(r_0) = -n$, and $b(r)/r = (r_0/r)^{n+1} \rightarrow 0$ at large radius.

The density and trace are

$$\rho = \frac{2n(\beta - 3\pi)b}{3(\beta - 8\pi)(\beta - 2\pi)r^3}, \quad (45)$$

$$\tau = -\frac{2n(\beta - 6\pi)b}{3(\beta - 8\pi)(\beta - 2\pi)r^3}. \quad (46)$$

while

$$\rho + p_r = \frac{(n+1)b}{(\beta - 8\pi)r^3}, \quad \rho + p_t = \frac{(n-1)b}{2(\beta - 8\pi)r^3}. \quad (47)$$

For $\beta > 8\pi$ and $n \geq 1$, the density is positive and both null combinations are non-negative for all $r \geq r_0$. The WEC and SEC are therefore satisfied throughout the spacetime. The limiting member $n = 1$ gives $b(r) = r_0^2/r$ and saturates the tangential NEC, $\rho + p_t \equiv 0$. The matter-geometry exchange is non-zero on this family; Eq. (15) reduces to

$$\nabla_\mu \Theta^\mu_r = -\frac{\beta n(n+3)r_0^{n+1}}{3(\beta - 8\pi)(\beta - 2\pi)r^{n+4}}, \quad (48)$$

which vanishes in the GR limit but not on the branch considered here.

Figure 1 illustrates the hierarchy of the energy conditions for representative members of the power-law family. Panel (a) confirms the asymptotically flat behaviour of the geometry. For $n = 2$, the density and both null combinations remain positive, but the radial DEC is violated. By contrast, $n = 6$ lies above the analytical threshold $n_{\text{DEC}} \simeq 4.782$ at $\beta = 40$ and satisfies all four standard

energy conditions. The numerical profiles are therefore consistent with the analytical bounds derived below.

The DEC gives one additional restriction. For $\beta > 8\pi$, the radial pressure is positive and the tangential pressure is negative. The tangential DEC therefore reduces to $\rho + p_t \geq 0$ and introduces no new bound. The radial condition $\rho - p_r \geq 0$ gives

$$n \geq n_{\text{DEC}}(\beta) = \frac{3(\beta - 2\pi)}{\beta - 6\pi}. \quad (49)$$

The threshold decreases from 9 as $\beta \rightarrow 8\pi^+$ and approaches 3 as $\beta \rightarrow \infty$. For $\beta = 40$, $n_{\text{DEC}} \simeq 4.782$. Equivalently,

$$\omega \equiv \frac{p_r}{\rho} = \frac{\beta(3+n) - 6\pi}{2n(\beta - 3\pi)}, \quad (50)$$

and the radial DEC is simply $\omega \leq 1$. Here ω is used only to characterise the reconstructed static source; no sound-speed or stability interpretation is attached to it.

Figure 2 summarises the analytical restrictions in the (B, n) plane. The line $B = 8$ separates the branch where the radial NEC necessarily fails at the throat from the non-exotic branch studied here. For $B > 8$, the region $n \geq 1$ satisfies the NEC, WEC, and SEC, whereas the stronger DEC requirement selects the region above $n_{\text{DEC}} = 3(B-2)/(B-6)$. The parameter map therefore provides a direct visual check of the bounds derived from the field equations.

It is useful to distinguish the physical matter from the effective source seen by the Einstein tensor. Writing $G^\mu_\nu = 8\pi\Theta^\mu_{\nu, \text{eff}}$, one obtains

$$\Theta^\mu_{\nu, \text{eff}} = \left(1 - \frac{\beta}{8\pi}\right) \Theta^\mu_\nu - \frac{\beta}{8\pi} \left(\mathcal{P} + \frac{\tau}{2}\right) \delta^\mu_\nu. \quad (51)$$

For either null direction,

$$(\rho + p_i)_{\text{eff}} = \left(1 - \frac{\beta}{8\pi}\right) (\rho + p_i), \quad i = r, t. \quad (52)$$

Since $\beta > 8\pi$, the physical and effective null combinations have opposite signs. For the power-law family,

$$\rho_{\text{eff}} = -\frac{nb}{8\pi r^3}, \quad p_{r, \text{eff}} = -\frac{b}{8\pi r^3}, \quad p_{t, \text{eff}} = \frac{(n+1)b}{16\pi r^3}, \quad (53)$$

and

$$(\rho + p_r)_{\text{eff}} = -\frac{(n+1)b}{8\pi r^3} < 0, \quad (54)$$

$$(\rho + p_t)_{\text{eff}} = -\frac{(n-1)b}{16\pi r^3} \leq 0. \quad (55)$$

The effective source retains the exotic behaviour required by the wormhole geometry, while the physical matter can satisfy the standard energy conditions. The trace coupling controls the mapping between these two descriptions.

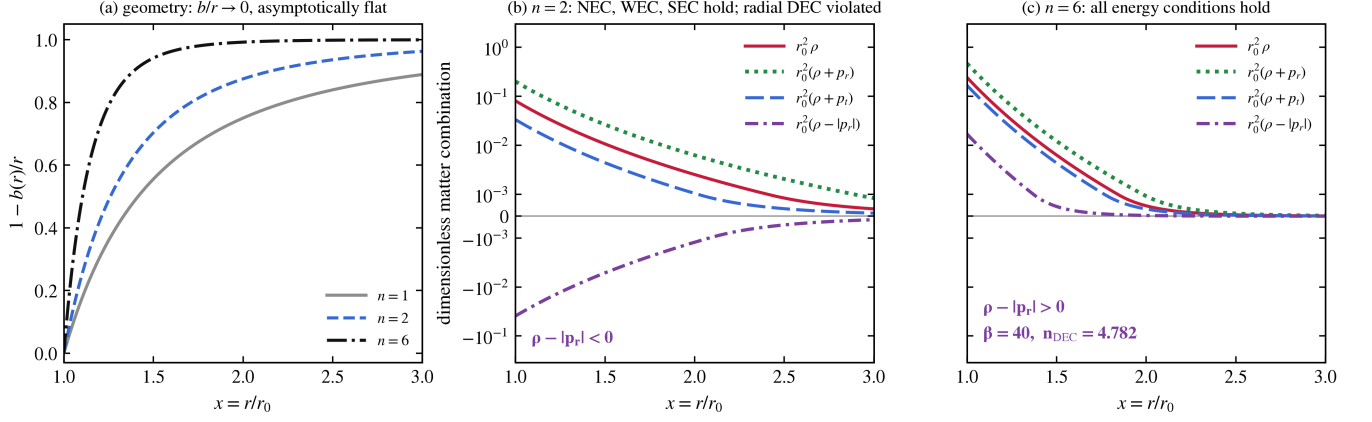


FIG. 1. Geometry and matter variables for $b(r) = r_0(r_0/r)^n$ at $\beta = 40$. Panel (a) shows $1 - b/r$ for representative values of n and confirms the asymptotically flat behaviour. Panel (b) corresponds to $n = 2$, for which the NEC, WEC, and SEC are satisfied but the radial DEC is violated. Panel (c) shows $n = 6$, where all four standard energy conditions are satisfied.

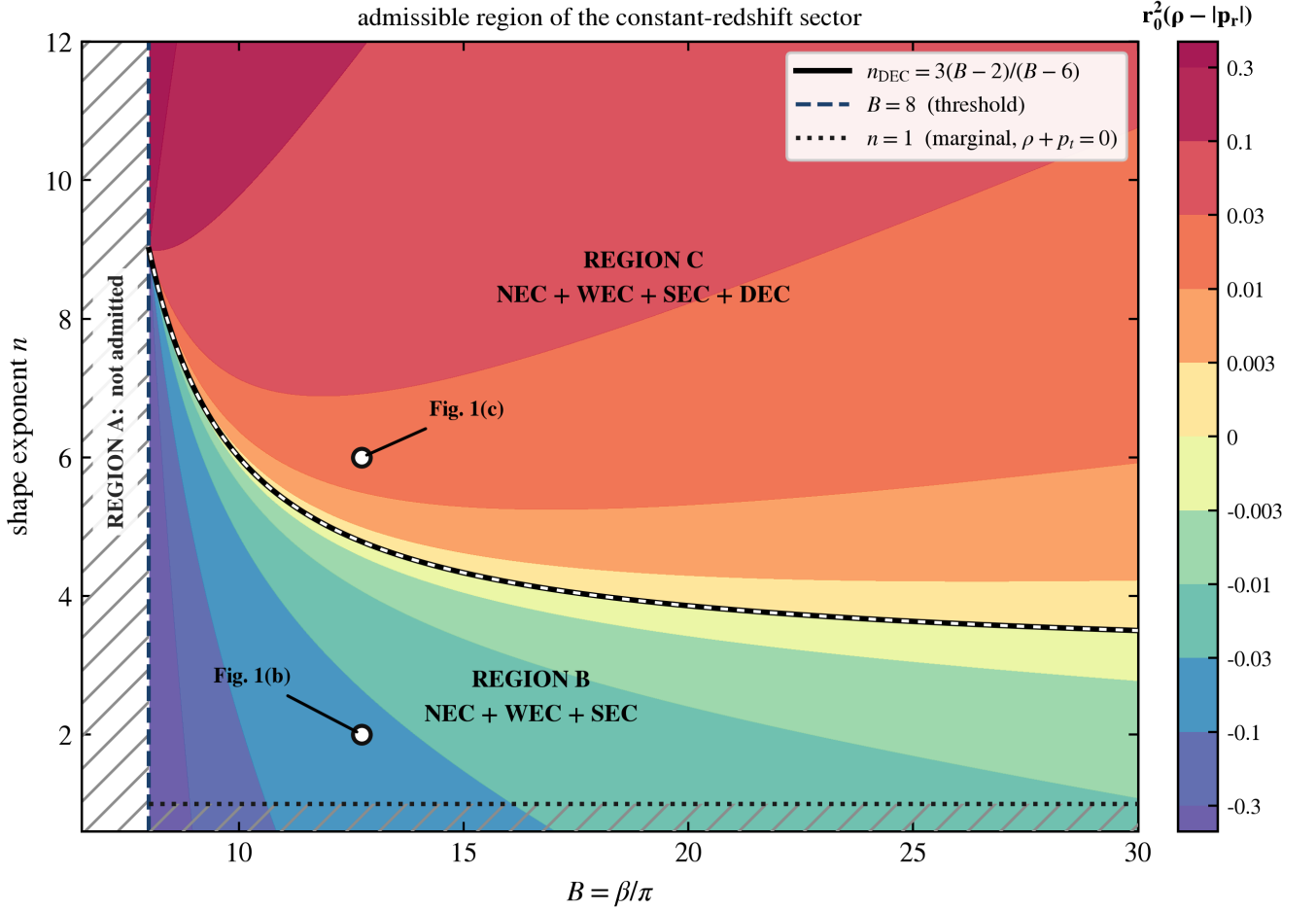


FIG. 2. Allowed region in the (B, n) plane, where $B = \beta/\pi$. The colour field is the dimensionless radial-DEC margin $r_0^2(\rho - |p_r|)$ evaluated at the throat $r = r_0$. The hatched region $B \leq 8$ is excluded by the radial NEC, while the hatched strip $n < 1$ for $B > 8$ lies outside the class satisfying the NEC, WEC, and SEC. For $B > 8$ and $n \geq 1$, the NEC, WEC, and SEC are satisfied, and above the curve $n_{\text{DEC}} = 3(B-2)/(B-6)$ the DEC is satisfied as well.

The same sign reversal also has a broader physical implication for the strong-coupling branch. For pressureless

matter, $p_r = p_t = 0$, one has $\mathcal{P} = 0$ and $\tau = -\rho$, so Eq. (51) gives

$$\rho_{\text{eff}} = \left(1 - \frac{3\beta}{16\pi}\right) \rho. \quad (56)$$

Thus positive-density dust has a negative effective density throughout the wormhole branch $\beta > 8\pi$. The non-exotic solutions should therefore be interpreted with this qualification: the coupling does not remove the exoticity required by the geometry, but relocates it to the effective gravity sector. This opposite-sign effective response also indicates that the weak-field viability of the $\beta > 8\pi$ branch is a separate question and cannot be inferred from the present wormhole analysis alone.

The behaviour shown in Fig. 3 makes the supporting mechanism transparent for the explicit choice $\beta = 40$ and $n = 5$. Both physical null-energy combinations remain positive, while their effective counterparts are negative. Since $1 - \beta/(8\pi) < 0$ on the branch $\beta > 8\pi$, this sign reversal follows directly from Eq. (52). The exoticity required by the wormhole geometry is therefore carried by the effective gravitational source rather than by the physical anisotropic matter.

The geometry is also regular at the throat. For the family (43),

$$R = -\frac{2nr_0^{n+1}}{r^{n+3}}, \quad (57)$$

$$R_{\mu\nu}R^{\mu\nu} = \frac{3n^2 + 2n + 3}{2} \frac{r_0^{2n+2}}{r^{2n+6}}, \quad (58)$$

$$K \equiv R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 2(n^2 + 2n + 3) \frac{r_0^{2n+2}}{r^{2n+6}}. \quad (59)$$

All three invariants remain finite at $r = r_0$ and vanish at infinity. The proper radial distance is

$$\ell(r) = \pm \int_{r_0}^r \frac{du}{\sqrt{1 - (r_0/u)^{n+1}}}, \quad (60)$$

which is finite at the throat. For $n = 1$ it reduces to $\ell(r) = \pm \sqrt{r^2 - r_0^2}$. In the constant-redshift sector the radial tidal term vanishes identically, while the remaining lateral tidal effect is finite for a finite throat radius. The solutions are therefore regular, horizon-free, and geometrically traversable.

IV. WORMHOLES WITH A NON-CONSTANT REDSHIFT FUNCTION

The member $n = 1$ lies exactly at the boundary of the allowed constant-redshift family because $\rho + p_t = 0$. This makes

$$b(r) = \frac{r_0^2}{r} \quad (61)$$

a natural case in which to ask whether a varying redshift function can turn the saturated tangential NEC into a strict inequality.

A. General behaviour of the null energy condition

Substitution of Eq. (61) into the general null-energy expressions gives

$$\rho + p_r = \frac{2[r_0^2 - r(r^2 - r_0^2)\Phi']}{(\beta - 8\pi)r^4}, \quad (62)$$

$$\rho + p_t = \frac{(r_0^2 - r^2)(\Phi'^2 + \Phi'') - r\Phi'}{(\beta - 8\pi)r^2}. \quad (63)$$

At the throat,

$$(\rho + p_r)|_{r_0} = \frac{2}{(\beta - 8\pi)r_0^2}, \quad (64)$$

while

$$(\rho + p_t)|_{r_0} = -\frac{\Phi'(r_0)}{(\beta - 8\pi)r_0}. \quad (65)$$

Thus, for $\beta > 8\pi$, a decreasing redshift function $\Phi'(r_0) < 0$ makes the tangential NEC strictly positive at the throat. The remaining question is whether this improvement can hold throughout the exterior.

Define

$$Y(r) = e^{\Phi(r)} > 0, \quad F(r) = \sqrt{r^2 - r_0^2} Y'(r). \quad (66)$$

Using $Y'' = (\Phi'' + \Phi'^2)Y$, Eq. (63) becomes

$$\rho + p_t = -\frac{\sqrt{r^2 - r_0^2}}{(\beta - 8\pi)r^2 Y(r)} F'(r). \quad (67)$$

For $\beta > 8\pi$ and $r > r_0$,

$$\rho + p_t \geq 0 \quad \Longleftrightarrow \quad F'(r) \leq 0. \quad (68)$$

At the throat $F(r_0) = 0$. For an asymptotically flat redshift function with $r\Phi'(r) \rightarrow 0$, one also has $F(\infty) = 0$. Therefore

$$\int_{r_0}^{\infty} F'(r) dr = F(\infty) - F(r_0) = 0. \quad (69)$$

If the tangential NEC were satisfied everywhere, then $F'(r) \leq 0$ over the whole exterior. Together with Eq. (69), this requires $F'(r) = 0$, hence $F(r) = 0$ and finally $\Phi'(r) = 0$. We therefore conclude that for the marginal shape function $b(r) = r_0^2/r$, an everywhere finite, asymptotically flat, non-constant redshift function cannot satisfy the tangential NEC throughout the entire exterior. A negative $\Phi'(r_0)$ can improve the condition locally at the throat, but the gain must be lost at some larger radius.

B. An illustrative redshift model

To illustrate the above result, consider

$$\Phi(r) = \frac{ar_0}{r}, \quad a > 0. \quad (70)$$

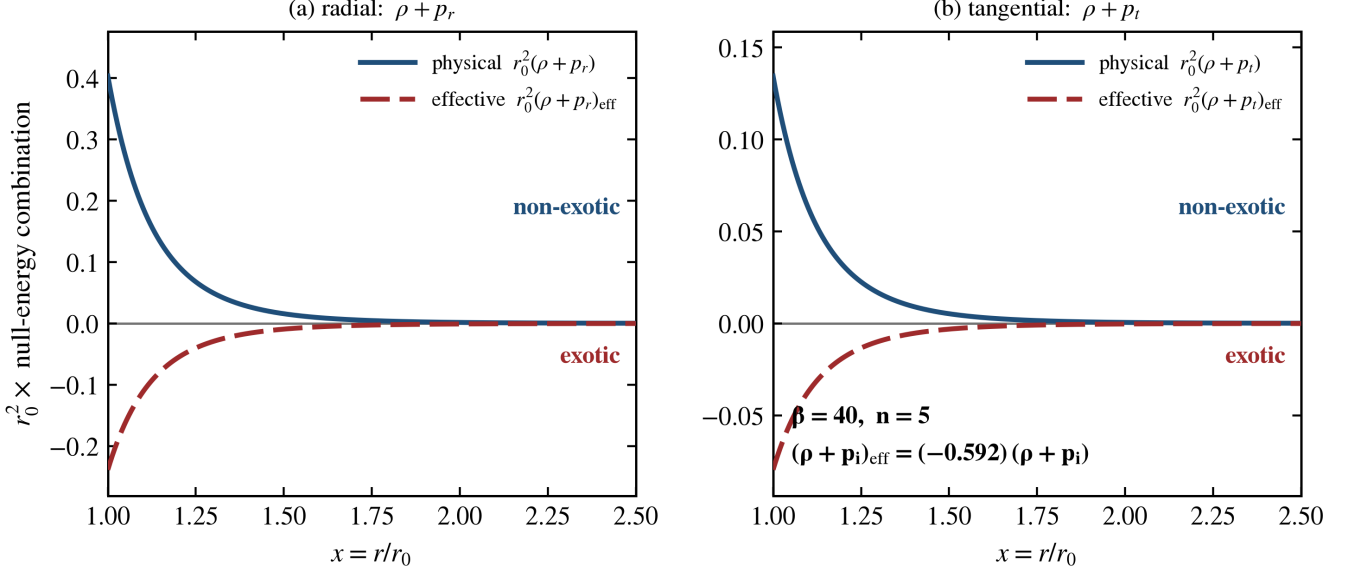


FIG. 3. Comparison between the physical and effective null-energy combinations for $\beta = 40$ and $n = 5$. The physical matter satisfies both null conditions, whereas the effective source violates them. The sign reversal follows directly from Eq. (52).

This function is finite everywhere, approaches zero at infinity, and satisfies $\Phi'(r_0) = -a/r_0 < 0$. The relevant matter variables are

$$\rho = \frac{r_0^2 [5a\beta r r_0 + 8(\beta - 3\pi)r^2 - 5a^2\beta(r^2 - r_0^2)]}{12(\beta - 8\pi)(\beta - 2\pi)r^6}, \quad (71)$$

$$\rho + p_r = \frac{2r_0 [a(r^2 - r_0^2) + r r_0]}{(\beta - 8\pi)r^5}, \quad (72)$$

$$\rho + p_t = -\frac{a r_0 [r(r^2 - 2r_0^2) + a r_0(r^2 - r_0^2)]}{(\beta - 8\pi)r^6}. \quad (73)$$

For $\beta > 8\pi$ and $a > 0$, the radial null combination is positive. The tangential combination changes sign at the root $x = r/r_0 > 1$ of

$$x^3 + ax^2 - 2x - a = 0. \quad (74)$$

The crossing occurs at $r/r_0 \simeq 1.358, 1.313, 1.247$, and 1.170 for $a = 1/4, 1/2, 1$, and 2 , respectively. Increasing the redshift gradient therefore improves the tangential NEC at the throat but causes it to fail closer to the throat outside.

A second restriction comes from the energy density. At large r its sign is controlled by $(8 - 5a^2)\beta - 24\pi$, giving the critical value

$$a_{\text{crit}}(\beta) = 2\sqrt{\frac{2}{5}}\sqrt{1 - \frac{3\pi}{\beta}}. \quad (75)$$

For $a > a_{\text{crit}}$, the density eventually becomes negative. At $\beta = 40$, $a_{\text{crit}} \simeq 1.106$, while for large β it approaches $2\sqrt{2/5} \simeq 1.265$.

Figure 4 displays the two limitations of the non-constant-redshift construction. For $a = 1/2 < a_{\text{crit}}$, the density and radial NEC remain positive, but the tangential NEC changes sign at $r \simeq 1.313 r_0$, showing explicitly that the throat-level improvement cannot persist throughout the exterior. For $a = 2 > a_{\text{crit}}$, the tangential NEC fails even closer to the throat and the density becomes negative at larger radius. The second panel therefore also illustrates the loss of the WEC when the redshift gradient is sufficiently large.

The explicit model confirms the general result. A non-constant redshift can improve the tangential NEC locally, but it does not convert the marginal $n = 1$ geometry into an everywhere strictly non-exotic solution. For sufficiently large redshift gradients, the WEC also fails because the energy density becomes negative.

V. SUMMARY AND CONCLUSIONS

We have investigated static and spherically symmetric traversable wormholes in the linear matter-coupled teleparallel model $f(T, \tau) = T + \beta\tau$, using an anisotropic source with the mean-pressure matter Lagrangian $\mathcal{L}_m = (p_r + 2p_t)/3$. The field equations were first obtained for a general Morris–Thorne geometry, so the role of the redshift function could be separated clearly from the conditions imposed by the throat. A key result is that the radial null energy condition at the throat depends on the coupling through the factor $\beta - 8\pi$. Since the flare-out condition makes the numerator positive, the branch that allows non-exotic physical matter is $\beta > 8\pi$. Importantly, this conclusion does not depend on $\Phi'(r_0)$.

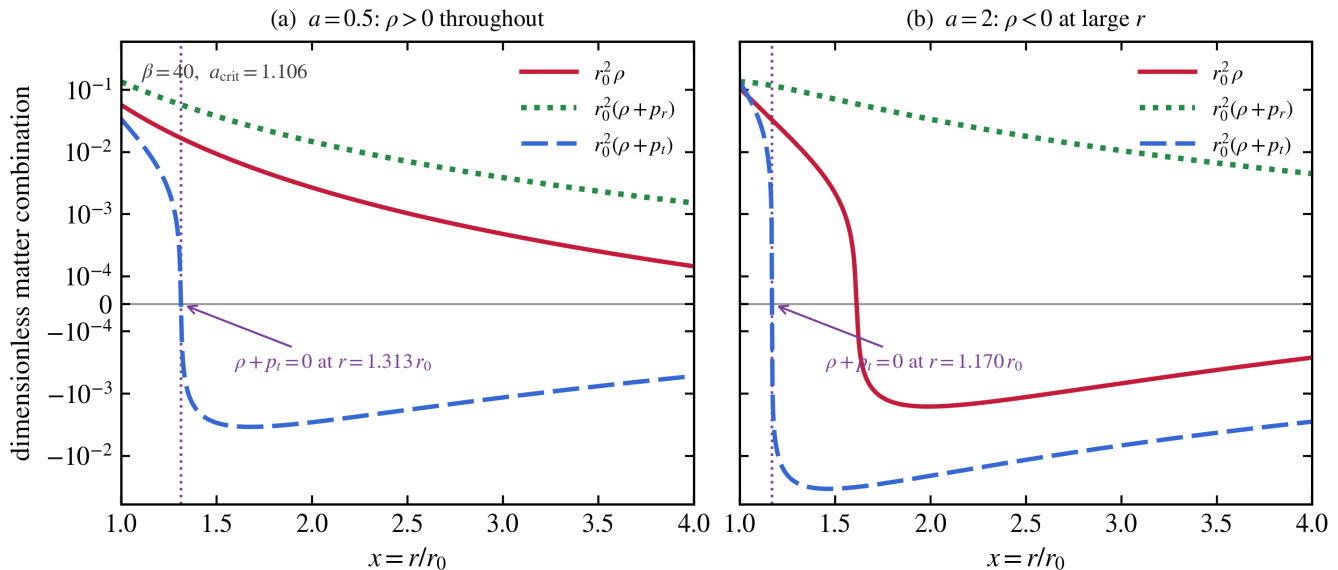


FIG. 4. Energy-condition behaviour for $b(r) = r_0^2/r$, $\Phi(r) = ar_0/r$, and $\beta = 40$. For $a = 1/2 < a_{\text{crit}}$, the density and radial NEC remain positive, whereas the tangential NEC changes sign at $r \simeq 1.313r_0$. For $a = 2 > a_{\text{crit}}$, the tangential NEC fails closer to the throat, at $r \simeq 1.170r_0$, and the density also becomes negative at larger radius.

The constant-redshift sector gives the clearest result of the analysis. For a positive shape function, the requirement that the physical matter satisfy the density condition together with the NEC, WEC, and SEC reduces to the single inequality $(rb)' \leq 0$. This condition does more than constrain the matter sector. It also forces the shape function to decrease sufficiently rapidly that the wormhole becomes asymptotically flat, remains below $b(r) = r$ outside the throat, and automatically satisfies the flare-out condition. In this sense, the same requirement that removes the need for exotic physical matter also selects a well-behaved wormhole geometry.

The admissible class can be represented by $b(r) = r_0^2 h(r)/r$, with $h(r_0) = 1$, $h > 0$, and $h' \leq 0$. The power-law choice $b(r) = r_0(r_0/r)^n$ provides a simple example. For this family, the NEC, WEC, and SEC are satisfied for $n \geq 1$, while the DEC introduces the stronger bound $n \geq 3(\beta - 2\pi)/(\beta - 6\pi)$. The corresponding solutions are regular at the throat, asymptotically flat, and traversable. The figures confirm these analytical results and show clearly how the dominant energy condition further restricts the allowed parameter space.

The distinction between the physical matter and the effective source is central to the interpretation. The wormhole geometry still requires an effective source that violates the null energy condition, as expected from the flare-out condition. However, for $\beta > 8\pi$ the physical and effective null-energy combinations have opposite signs. The same branch also gives an opposite-sign effective density for pressureless matter. Thus the trace coupling shifts the exotic behaviour required by the geometry into the effective gravitational sector rather than eliminating it. This makes the solutions mathematically

consistent but also shows that the weak-field viability of the strong-coupling branch must be assessed separately. The physical matter is likewise not separately conserved for $\beta \neq 0$, reflecting an exchange of energy-momentum with the gravitational sector.

A related global feature is that the complete admissible constant-redshift class has zero ADM mass. Although the local physical energy density can be positive, the bound imposed by the energy conditions forces $b(r) \rightarrow 0$ at infinity. This provides a useful asymptotic counterpart of the effective-source interpretation above.

We also examined the marginal member $b(r) = r_0^2/r$, which saturates the tangential NEC in the constant-redshift case. A decreasing redshift function can make the tangential NEC strictly positive at the throat, but the global analysis shows that this local improvement cannot be maintained throughout an asymptotically flat exterior unless the redshift function is constant. The explicit choice $\Phi = ar_0/r$ illustrates the same obstruction and shows that sufficiently large redshift gradients can also drive the energy density negative. Thus the additional freedom in $\Phi(r)$ does not rescue the marginal $n = 1$ configuration.

The present non-constant-redshift result is restricted to this boundary member of the power-law family. It remains to determine whether solutions with $n > 1$ and a varying redshift function can preserve the energy conditions globally. The dynamical stability of the non-exotic configurations is another important open question. Because the theory contains a non-minimal matter-torsion coupling, a consistent perturbative treatment should be derived within $f(T, \tau)$ gravity rather than imported directly from general relativity. These issues provide nat-

ural directions for extending the present analysis.

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